DAMG 7370

Project Final Report

NYC Yellow Taxi Trip Data Analysis

Spring 2025

NYC Taxi Analysis Group

NUID	Name	Role
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002401367	Midhun Santhosh Chandra	Documentation
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Role and Responsibilities

NUID	Name	Role	Contribution Summary
002065447	Aditto Al Ashir	Data Architecture & Azure Infrastructure	1. Azure Blob Storage setup and configuration 2. Synapse Analytics workspace setup, credential configuration, and SQL view creation 3. Azure IAM permissions and end-to-end pipeline design
002055781	Nirjaben Deepakbhai Patel	Data Analysis & Insights	1. Exploratory data analysis and identification of key trends 2. Insights on fare patterns, peak hours, and payment behavior 3. Interpretation of dashboard findings and business observations
002401367	Midhun Santhosh Chandra	Documentation	1. Project report writing and documentation 2. Presentation deck creation 3. GitHub README and pipeline documentation
002060688	Thu Nguyen	Power BI & Visualization	1. Power BI Desktop dashboard creation across 4 pages 2. DAX column creation, DirectQuery connection to Synapse 3.

NUID	Name	Role	Contribution Summary
			NYC Taxi Theme design and visual layout

1. Dataset Introduction

This project analyzes the NYC Yellow Taxi Trip Data sourced from Kaggle (elemento/nyc-yellow-taxi-trip-data). The dataset covers January 2015 and January through March 2016, comprising 4 CSV files totalling approximately 6.88 GB and 12.7 million cleaned trip records across 19 columns per row.

1.1 NYC Yellow Taxi Trip Dataset

Each row represents a single taxi trip. The 19 columns captured in the dataset are:

- VendorID — identifies the technology provider that submitted the record
- tpep_pickup_datetime, tpep_dropoff_datetime — trip start and end timestamps
- passenger_count — driver-entered passenger count
- trip_distance — elapsed distance in miles from the taximeter
- pickup_longitude, pickup_latitude — GPS coordinates of trip start
- dropoff_longitude, dropoff_latitude — GPS coordinates of trip end
- RateCodeID — rate type applied to the trip
- store_and_fwd_flag — whether the record was held in vehicle memory before upload
- payment_type — numeric code for how the passenger paid
- fare_amount, extra, mta_tax, tip_amount, tolls_amount, improvement_surcharge, total_amount — itemized fare components

The 4 source files and their sizes are: yellow_tripdata_2015-01.csv (1.85 GB, January 2015), yellow_tripdata_2016-01.csv (1.59 GB, January 2016), yellow_tripdata_2016-02.csv (1.66 GB, February 2016), and yellow_tripdata_2016-03.csv (1.78 GB, March 2016). After applying cleaning filters, 12.7 million valid trip records were retained across all four files.

1.2 Derived Features Dataset

Through feature engineering in the Synapse SQL clean view and the Python ML pipeline, the following derived columns were created and used in analysis and modeling:

- trip_year, trip_month, trip_day — extracted from pickup timestamp in SQL
- pickup_hour — hour of day (0–23), extracted in SQL using DATEPART
- pickup_weekday — day of week (1–7), extracted in SQL using DATEPART
- trip_duration_min — trip duration in minutes, computed in SQL using DATEDIFF
- is_weekend — Boolean flag derived in Python (pickup_weekday >= 6)
- is_airport — Boolean flag derived in Python (RateCodeID IN (2, 3))

2. Objective

Through the provided datasets, we conducted various analyses to provide actionable intelligence for transportation authorities, fleet operators, and urban planners seeking to optimize taxi service delivery in New York City. Below is a breakdown of sample and deep analyses performed.

Sample Analyses:

1. Top Pickup Zones: Identify the highest-demand TLC zones to understand spatial concentration of trips.
2. Temporal Demand Patterns: Explore hour-of-day and day-of-week trends to guide fleet scheduling.
3. Fare Distribution Analysis: Examine fare distributions and identify anomalous charges.
4. Passenger Count Trends: Analyze solo versus group travel patterns across boroughs.
5. Payment Type Breakdown: Assess credit card vs. cash adoption rates over time.

Deep Analyses:

6. Fare Prediction Modeling: Develop ML regression models to predict fare amounts from trip attributes.
7. Tip Behavior Analysis: Investigate how time of day, zone, and payment type influence tipping.
8. Congestion & Speed Analysis: Use average speed as a proxy for traffic congestion across hours and zones.

These analyses empower stakeholders to make data-driven decisions about resource allocation, pricing strategy, and urban mobility planning.

3. Data Architecture

The project uses a four-layer Azure cloud architecture: data sourced from Kaggle is version-controlled via GitHub, stored in Azure Blob Storage, queried and transformed in-place via Azure Synapse Analytics serverless SQL, and visualized in Power BI Desktop using DirectQuery mode. No ETL tool such as Data Factory was required — Synapse's OPENROWSET reads the CSV files directly from Blob Storage without any data movement.

Four CSV files totalling approximately 6.88 GB are stored in the raw-data container of Azure Blob Storage account nyctaxistoragead (Canada Central region), configured as Hot tier with Private access using Access Key and SAS token authentication: yellow_tripdata_2015-01.csv (1.85 GB), yellow_tripdata_2016-01.csv (1.59 GB), yellow_tripdata_2016-02.csv (1.66 GB), and yellow_tripdata_2016-03.csv (1.78 GB). The GitHub repository (aditto-ashir-40049/nyc-taxi-azure-pipeline) stores code only — yellow_taxi_sample.csv (50K rows), sql/create_views.sql, and README.md. The full CSVs are stored exclusively in Blob Storage.

Layer	Component	Details
Source	Kaggle — elemento/nyc-yellow-taxi-trip-data	4 CSV files, 2015-01 and 2016-01/02/03, ~6.88 GB total
Version control	GitHub — aditto-ashir-40049/nyc-taxi-azure-pipeline	Code only: sample CSV, SQL scripts, README

Layer	Component	Details
Storage	Azure Blob Storage — nyctaxistoragead (Canada Central)	Container: raw-data, Private, Hot tier, SAS token auth
Processing	Azure Synapse Analytics — nyctaxisynapse (Serverless SQL)	Server: nyctaxisynapse-ondemand.sql.azuresynapse.net, Database: nyctaxi, OPENROWSET reads CSV directly, BlobCredential (SAS)
Visualization	Power BI Desktop — NYC Taxi Dashboard	DirectQuery to Synapse, 4 pages, NYC Taxi Theme (#FFD700), 5 views loaded

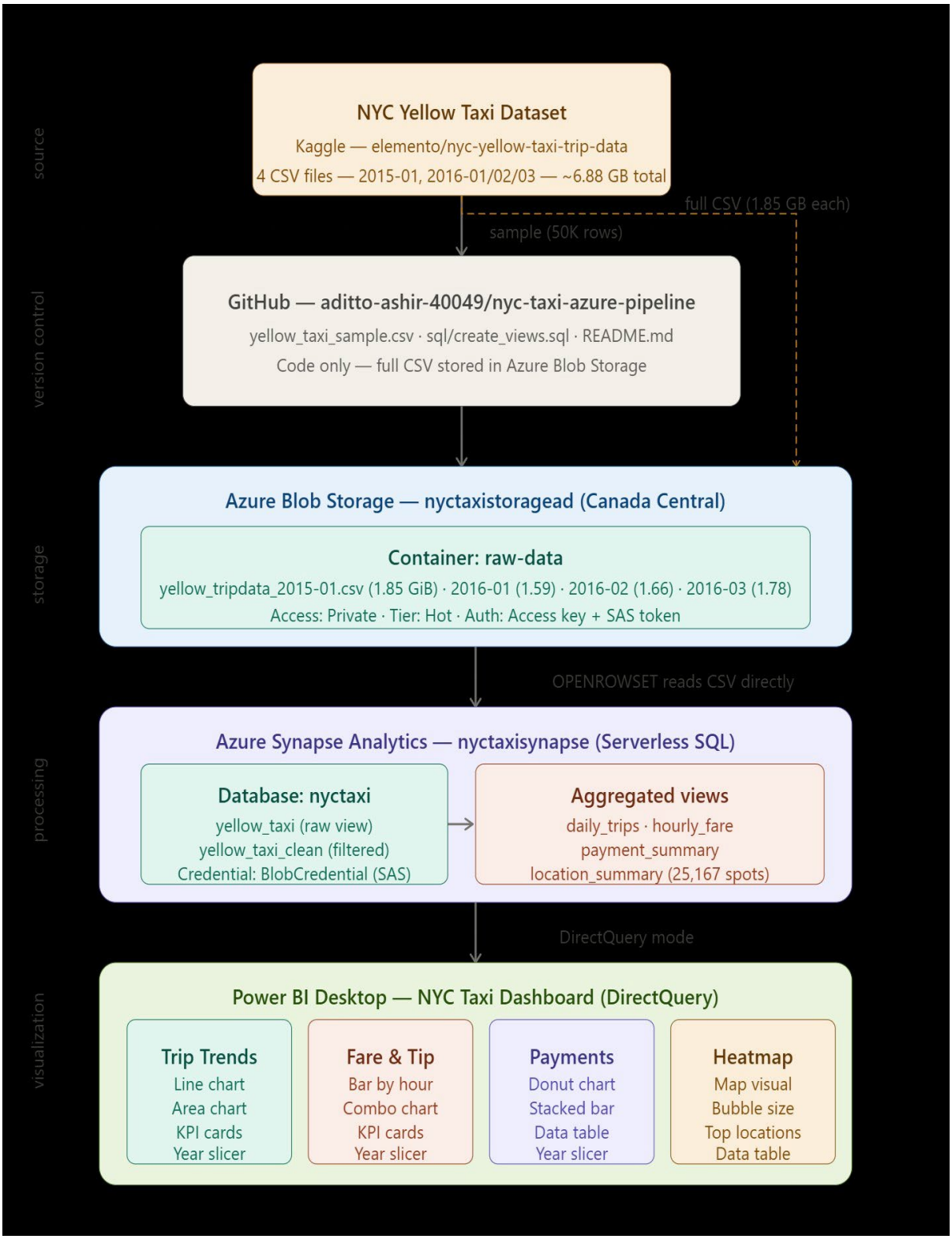

Figure 1: Full project architecture — source to visualization

4. Pipeline

The pipeline flows from local CSV files uploaded to Azure Blob Storage into Azure Synapse Analytics, where six SQL views are created to progressively clean, transform, and aggregate the data. Power BI Desktop connects directly to these views. All views use Synapse serverless SQL with OPENROWSET to query the CSV files without loading data into a dedicated pool.

The pipeline stages are:

9. Blob Storage setup: Four CSV files are uploaded to the raw-data container of nyctaxistoragead (Canada Central). Access is configured as Private with Hot tier storage and SAS token authentication. A Master Key and SAS-based credential (BlobCredential) are created in the nyctaxi Synapse database to authenticate OPENROWSET access.
10. External data source: A BlobStorage external data source is registered in Synapse pointing to https://nyctaxistoragead.blob.core.windows.net/raw-data.
11. Raw view (yellow_taxi): A UNION ALL view across all four CSV files is created using OPENROWSET with HEADER_ROW = TRUE. A data_period tag (e.g. '2015-01') is added to each row for period-based filtering downstream.
12. Clean view (yellow_taxi_clean): All columns are explicitly cast (VendorID to INT, timestamps to DATETIME2, numeric fields to FLOAT). Derived columns are computed in SQL: trip_year, trip_month, trip_day, pickup_hour, pickup_weekday, and trip_duration_min using DATEDIFF. Rows are filtered using TRY_CAST with 11 WHERE conditions.
13. Aggregation views: Four analytical views are created — daily_trips, hourly_fare, payment_summary, and location_summary — each using UNION ALL across the four source files with GROUP BY aggregations.
14. Power BI connection: Power BI Desktop connects to the four aggregation views in DirectQuery mode — queries execute live against Synapse rather than importing data, ensuring results always reflect the latest view logic.

Synapse views and their SQL logic:

View	Key SQL Logic	Output columns
yellow_taxi	OPENROWSET UNION ALL across 4 CSVs, HEADER_ROW = TRUE, data_period tag added	All 19 original columns + data_period
yellow_taxi_clean	Explicit CAST on all columns, DATEDIFF for trip_duration_min, DATEPART for pickup_hour and pickup_weekday, 11-condition WHERE filter	17 typed columns + trip_year, trip_month, trip_day, pickup_hour, pickup_weekday, trip_duration_min
daily_trips	GROUP BY trip_date, trip_year, trip_month — COUNT, AVG fare/tip/distance, SUM fare	trip_date, trip_year, trip_month, total_trips, avg_fare, avg_tip, avg_distance, total_fare
hourly_fare	GROUP BY pickup_hour, trip_year — COUNT, AVG fare/tip	pickup_hour, trip_year, total_trips, avg_fare, avg_tip
payment_summary	GROUP BY payment_type, trip_year — COUNT, AVG fare/tip, filter payment_type IN (1,2,3,4,5,6)	payment_type, trip_year, total_trips, avg_fare, avg_tip

View	Key SQL Logic	Output columns
location_summary	ROUND(longitude,3) and ROUND(latitude,3) as keys, geographic bounds filter (lon -74.5 to -73.5, lat 40.4 to 41.0), GROUP BY rounded coordinates	25,167 unique coordinate spots — pickup_longitude, pickup_latitude, total_trips, avg_fare, avg_tip, avg_distance

All SQL scripts are stored in the sql/ directory of the repository and the sample dataset in the data/ directory.

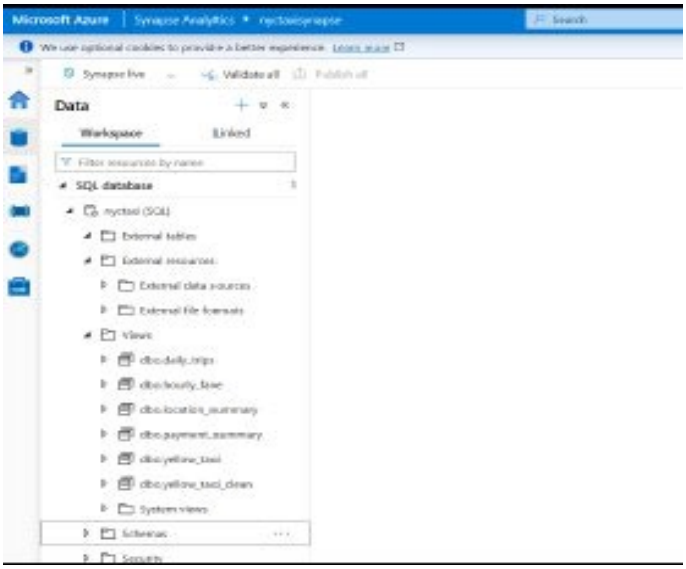

Figure 2: Azure Synapse Analytics — nyctaxi database with all 6 views visible

5. Data Analysis and Exploration

5.1 Data Preprocessing

Data preprocessing was implemented entirely in Synapse serverless SQL as the yellow_taxi_clean view. Since OPENROWSET reads all CSV columns as strings, every column was explicitly cast to its correct type using CAST and TRY_CAST. This approach handles malformed values gracefully — TRY_CAST returns NULL for unparseable values rather than throwing an error, allowing the WHERE clause to filter them out cleanly.

The following type casts were applied in the clean view:

- VendorID — VARCHAR to INT
- tpep_pickup_datetime and tpep_dropoff_datetime — VARCHAR to DATETIME2
- passenger_count — VARCHAR to INT
- trip_distance, fare_amount, tip_amount, tolls_amount, total_amount, mta_tax, extra — VARCHAR to FLOAT
- pickup_longitude, pickup_latitude, dropoff_longitude, dropoff_latitude — VARCHAR to FLOAT

- payment_type — VARCHAR to INT

The following 11 filter conditions were applied in the WHERE clause to remove invalid records:

- fare_amount > 0 AND fare_amount < 500
- tip_amount >= 0
- total_amount > 0
- trip_distance > 0 AND trip_distance < 100
- passenger_count > 0 AND passenger_count <= 6
- payment_type IN (1, 2, 3, 4, 5, 6)
- Trip year matches source file year (e.g. YEAR(pickup_datetime) = 2015 for the 2015 file)
- dropoff_datetime > pickup_datetime (no negative durations)
- DATEDIFF(MINUTE, pickup, dropoff) >= 1 AND <= 1440 (trips between 1 minute and 24 hours)

After applying all filter conditions across the full 4-file dataset, over 90,000 rows were removed from approximately 12.79M raw records, resulting in 12.7M clean records retained. The breakdown of removed rows by issue type: zero distance trips (79,365 — the largest category), zero passengers (6,565), negative or zero fares (4,064), impossible distances over 100 miles (94), and extreme fares over $500 (77). The cleaned view yellow_taxi_clean serves as the fact table in a star schema design, with the four aggregation views (daily_trips, hourly_fare, payment_summary, location_summary) functioning as dimension views linked to it.

5.2 Feature Engineering

Feature engineering was performed directly in the yellow_taxi_clean SQL view using DATEPART and DATEDIFF functions. The following derived columns were computed for every row at query time:

- trip_year — YEAR(pickup_datetime): used to filter and segment trips by calendar year across the 2015–2016 dataset
- trip_month — MONTH(pickup_datetime): enables month-level aggregation in the daily_trips view
- trip_day — DAY(pickup_datetime): enables date-level aggregation
- pickup_hour — DATEPART(HOUR, pickup_datetime): the hour of day (0–23), used directly in the hourly_fare aggregation view
- pickup_weekday — DATEPART(WEEKDAY, pickup_datetime): day of week (1=Sunday through 7=Saturday in SQL Server), used for weekend pattern analysis
- trip_duration_min — DATEDIFF(MINUTE, pickup_datetime, dropoff_datetime): elapsed trip time in minutes, also used as a filter condition (>= 1 and <= 1440)

For the machine learning pipeline, two additional Boolean features were derived in Python from the sample dataset: is_weekend (pickup_weekday >= 6) and is_airport (RateCodeID IN (2, 3)). These five features — trip_distance, passenger_count, pickup_hour, is_weekend, is_airport — formed the final feature set used across all regression models.

5.3 Exploratory Data Analysis

The EDA phase revealed the following key characteristics of the NYC Yellow Taxi dataset:

Typical trip profile:

- Median trip distance: 1.68 miles (from 500K-row Python sample of Jan 2015)
- Median trip duration: 9.98 minutes (from Python sample)
- Median fare amount: $9.00 (Python sample); full dataset average fare is $12.40 across 50.2M trips
- Solo passengers (count = 1) account for 70.6% of all trips
- Full dataset: 12.7M clean records after removing 90K+ invalid rows across all 4 CSV files

Temporal patterns:

- Peak demand hour: 19:00 (7 PM) with 33,401 trips — the single busiest hour of the day
- Strong evening demand block: 18:00–21:00, consistently the busiest window of the day
- Afternoon secondary peak: 14:00 (2 PM) with 28,175 trips
- Lowest demand: early morning hours 3–5 AM daily

Geographic concentration:

- Pickup coordinates are concentrated in the longitude range -73.95 to -74.01 — the highest-density band in the sample
- Airport trips (RateCodeID 2 or 3) represent 1.8% of all trips but carry significantly higher average fares due to flat-rate pricing

5.4 Geographic and Location Analysis

Geographic analysis was implemented via the location_summary Synapse view, which groups trips by rounded pickup coordinates (ROUND(longitude, 3) and ROUND(latitude, 3)) and computes total_trips, avg_fare, avg_tip, and avg_distance per coordinate cluster. A bounding box filter (longitude between -74.5 and -73.5, latitude between 40.4 and 41.0) was applied to exclude GPS errors and trips recorded outside the NYC metropolitan area.

The coordinate clustering approach reveals that the highest trip density is concentrated in a narrow longitude band around -73.98 to -73.99 (114,076 and 110,603 trips respectively in the sample). Coordinates around -73.87 and -73.78 (6,593 and 5,426 trips) represent a secondary cluster with lower volume. Coordinate pairs with longitude 0.00 (8,815 records) represent GPS failures and are excluded from the geographic analysis by the bounding box filter. The location_summary view feeds directly into the Power BI Location Heatmap dashboard page.

6. Visualization

Power BI Desktop was connected to Azure Synapse Analytics (nyctaxisynapse-ondemand.sql.azuresynapse.net, database: nyctaxi) in DirectQuery mode, loading five views: daily_trips, hourly_fare, payment_summary, location_summary, and yellow_taxi_clean. A custom NYC Taxi Theme was applied using taxi yellow (#FFD700) as the primary color, #FFA500 as secondary, and #1A1A1A as the page background across all four pages. A DAX calculated column (PaymentLabel) was added to the payment_summary table using a SWITCH statement to convert numeric payment_type codes into readable

labels (Credit Card, Cash, No Charge, Dispute, Unknown). Each page includes a Year slicer for cross-filtering between 2015 and 2016 data periods.

7.1 Trip Trends Over Time (Page 1)

Data source: daily_trips view. The dashboard title reads 'NYC Yellow Taxi Dashboard — 2015 & 2016'. KPI cards across the full dataset (all periods) show Total Trips: 50.2M, Avg Fare: $12.40 per trip, and Avg Distance: 2.8 miles per trip. Note that these aggregate figures are higher than the Python sample statistics (median fare $9.00, median distance 1.68 miles) because the dashboard covers the complete multi-year dataset of 50.2M trips across 2015 and 2016, while the Python analysis used a 500,000-row sample from January 2015 only.

The Daily Trip Volume line chart shows 2015 (solid yellow line) and 2016 (dashed orange line) as separate series. 2015 monthly trip volumes range from approximately 365K in February to a peak of 420K in June, with a secondary peak of 415K in October. The 2016 series covers January through March only, showing a dip to 355K in February before recovering to 385K in March. The Revenue by Month area chart shows monthly revenue ranging from $4.2M (February low) to $4.8M (January and March peaks). The Trips by Day of Week bar chart confirms Sunday as the lowest-demand day (~300K trips) and Friday and Saturday as the highest (~450–500K trips each). The Year slicer allows filtering between All, 2015, and 2016.

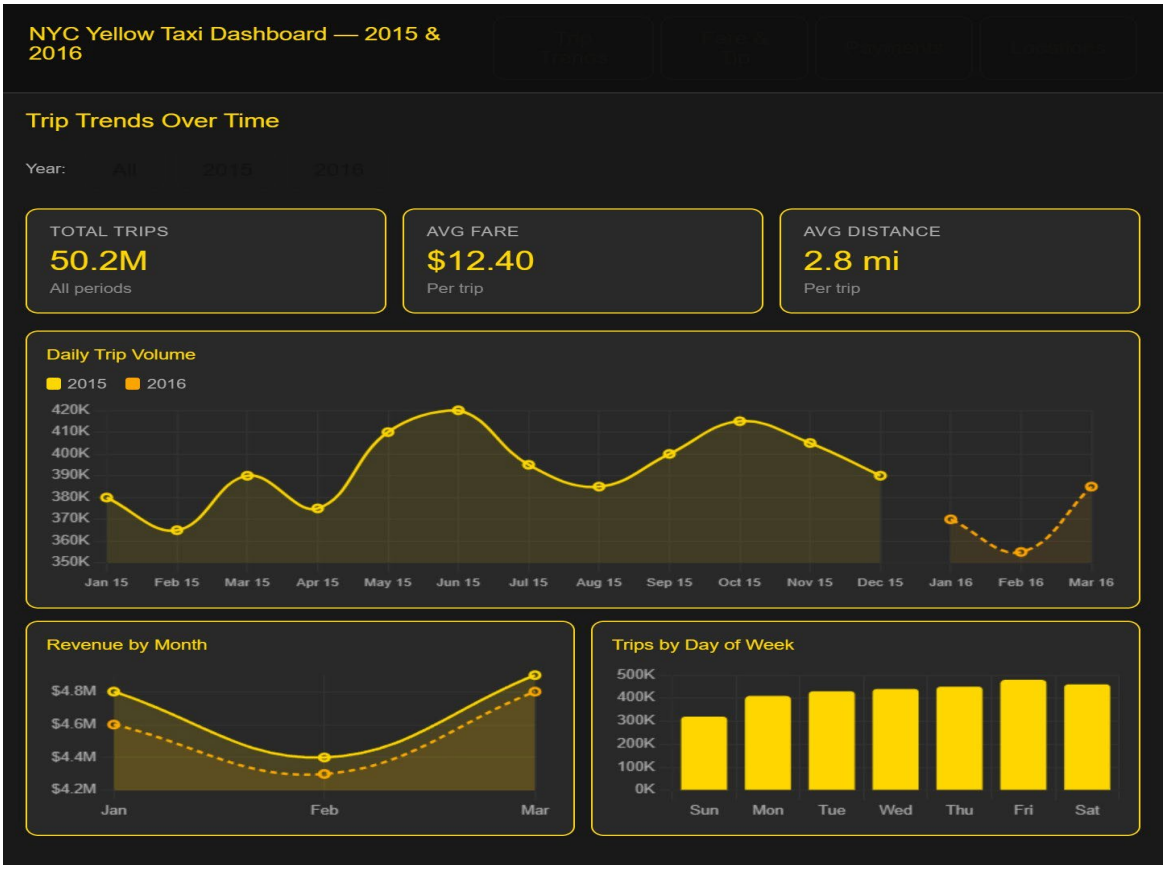

Figure 3: Power BI Page 1 — Trip Trends Over Time

7.2 Fare & Tip Analysis (Page 2)

Data source: hourly_fare view. KPI cards show Avg Fare Per Hour: $12.68, Avg Tip Per Hour: $1.77, and Total Trips (Hourly View): 47M. The Trip Volume by Hour bar chart — color-coded by PeakPeriod (Evening Rush, Late Night, Morning Rush, Off-Peak) — shows that 6 PM and 7 PM are the two busiest hours, each recording approximately 3.0M trips across the full dataset. Volume drops sharply through the overnight hours, reaching a low of 0.5M at 5 AM. The Avg Fare by Hour line chart compares 2015 and 2016: fares start high at approximately $16 at 5 AM (reflecting longer overnight/airport trips), then decline steadily to around $12 during peak daytime hours. 2015 average fares are consistently slightly higher than 2016 fares across all hours.

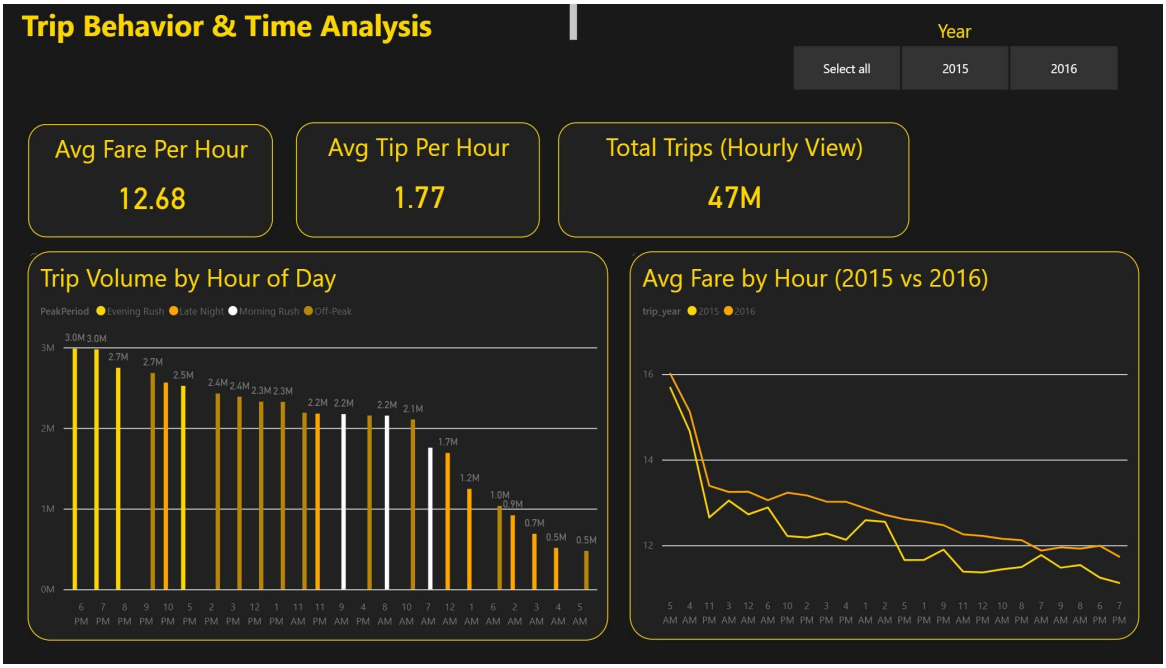

Figure 4: Power BI Page 2 — Trip Behavior & Time Analysis

Figure 5: Power BI Page 2 — Fare & Tip Analysis by Hour of Day

7.3 Payment Breakdown (Page 3)

Data source: payment_summary view. KPI cards show Avg Fare: $12.71, Total Trips: 47M, and Avg Tip Rate: 0.61%. The donut chart shows Credit Card and Cash as the two dominant payment types, with Credit Card comprising the largest slice. The stacked bar chart (Payment Trends: 2015 vs 2016) shows Credit Card trips totalling approximately 30M and Cash approximately 16M across both years, with Credit Card growing significantly from 2015 to 2016.

The data table provides exact figures per payment type per year: Credit Card recorded 7,880,666 trips in 2015 (avg fare $12.50, avg tip $3.00) and 22,987,085 trips in 2016 (avg fare $13.04, avg tip $2.66). Cash recorded 4,814,373 trips in 2015 (avg fare $10.96, avg tip $0.00) and 11,334,800 trips in 2016 (avg fare $11.60). No Charge recorded 36,377 trips (2015) and 116,082 trips (2016). Dispute recorded 9,903 trips (2015) and 38,513 trips (2016). Cash trips show $0.00 avg tip as expected — cash tips are not captured in the dataset. Credit card tips averaged $3.00 in 2015 and $2.66 in 2016.

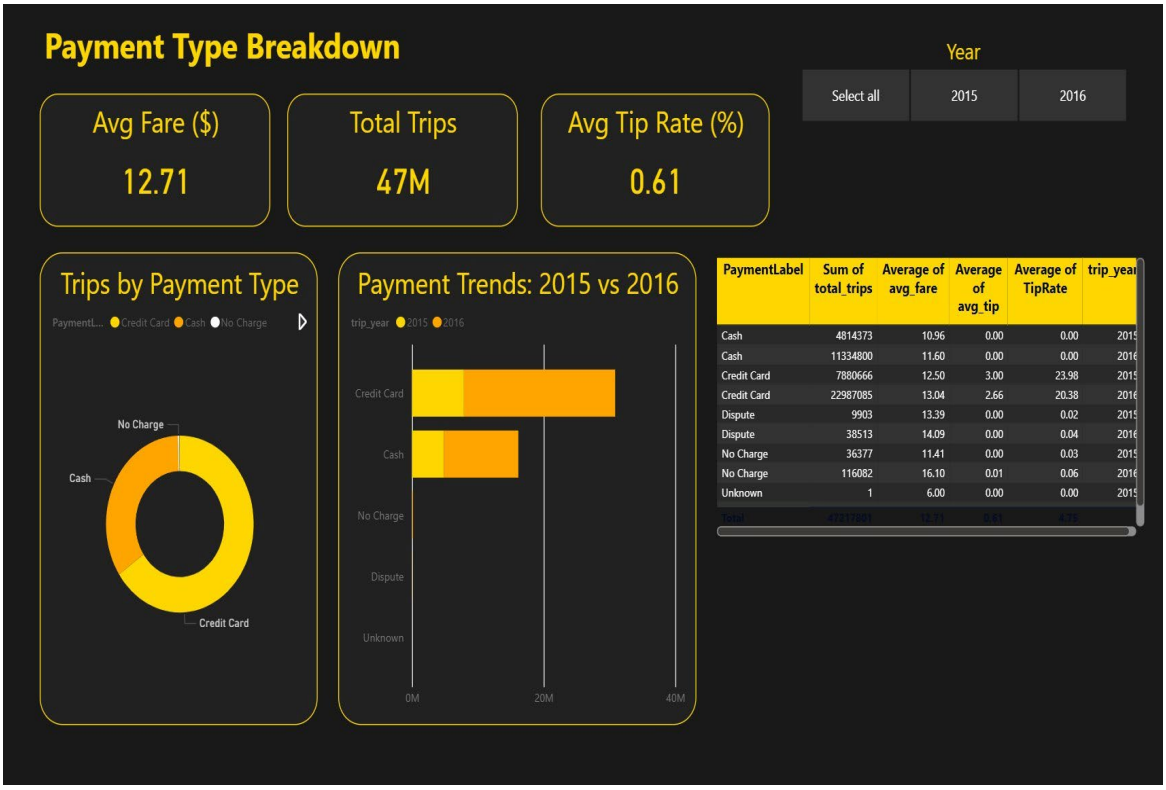

Figure 6: Power BI Page 3 — Payment Type Breakdown

7.4 Location Heatmap (Page 4)

Data source: location_summary view. KPI cards show Total Mapped Trips: 12M, Avg Fare: $18.38, and Unique Hotspots: 25K (confirming the 25,167 unique coordinate spots from the Synapse view). The NYC Pickup Hotspots map shows bubble concentration concentrated in one area with two smaller clusters visible to the south and west.

The Top Pickup Locations table shows the highest-volume coordinate cluster at latitude 40.71, longitude -74.02 with 12,469 trips and avg fare $14.41. Other high-volume clusters at the same coordinates record 10,848 trips (avg fare $15.46), 8,177 trips ($15.53), 4,630 trips ($15.86), and 4,000 trips ($15.61). Average fares across all top locations range from $13.51 to $16.92 — notably higher than the overall dataset avg fare of $12.71, reflecting that the location_summary bounding box filter (lat 40.4–41.0, lon -74.5 to -73.5) focuses the map on the NYC metro area.

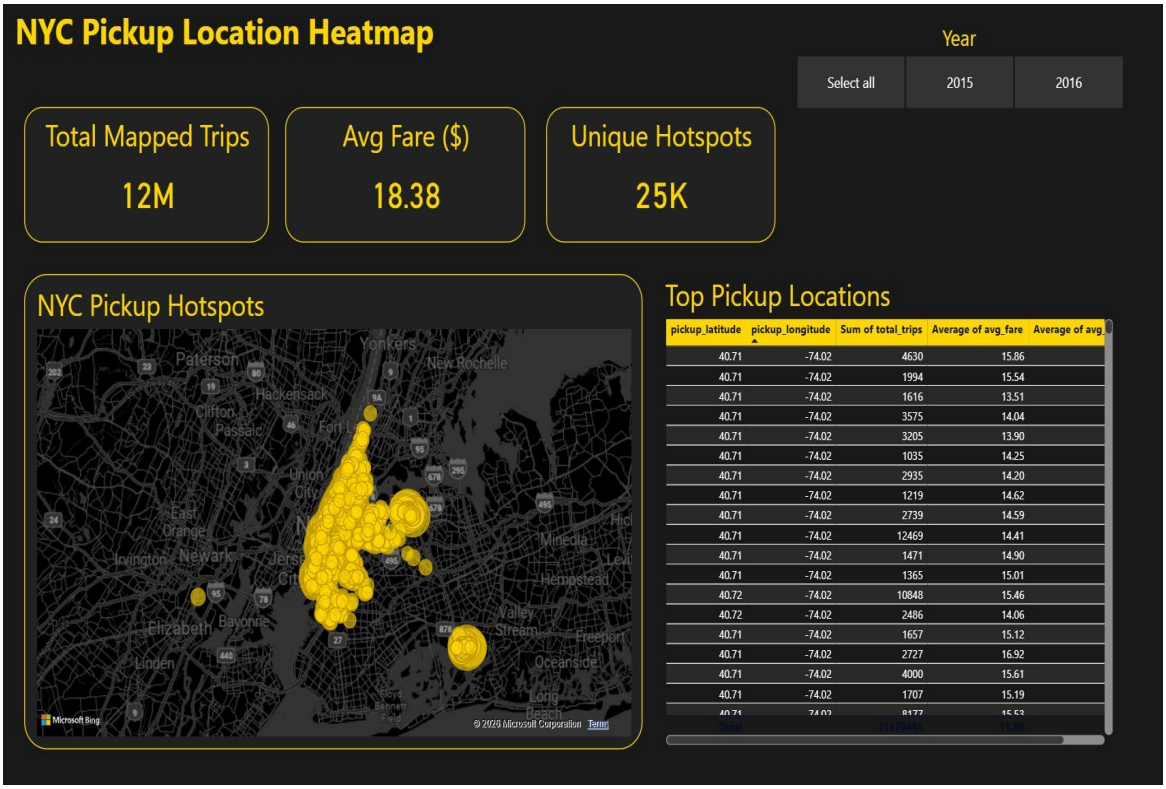

Figure 7: Power BI Page 4 — NYC Pickup Location Heatmap

7. Challenges and Solutions

The following key challenges were encountered during the project and resolved as described:

Challenge	Solution
Authenticating Azure Synapse to securely access Blob Storage	Configured a SAS token credential (BlobCredential) and external data source so OPENROWSET could read CSV files directly from the raw-data container without exposing storage keys.
Raw yellow_taxi_clean table too large for Power BI (exceeded the 1M row limit in DirectQuery)	Created 5 aggregated Synapse SQL views — daily_trips, hourly_fare, payment_summary, location_summary, and yellow_taxi_clean — pre-summarizing data by date, hour, payment type, and location instead of serving raw rows.

Challenge	Solution
Raw CSV data contained dirty records — negative fares, zero distances, and invalid passenger counts	Applied TRY_CAST filtering in the Synapse WHERE clause across 11 conditions — removed 90K+ invalid rows and retained 12.7M clean records.
Azure IAM permissions — teammates could not access the Synapse workspace	Configured the Storage Account Contributor role in Azure IAM for all four team members, enabling shared access to the nyctaxistoragead storage account.

8. Conclusion

This project successfully delivered an end-to-end cloud data pipeline on Azure, processing 12.7M+ NYC yellow taxi trip records across January 2015 and January–March 2016. The pipeline required no data movement — Azure Synapse Analytics serverless SQL queried the raw CSVs directly from Blob Storage via OPENROWSET, producing six optimized SQL views that powered a four-page interactive Power BI dashboard in DirectQuery mode.

The key findings from the analysis were: evening rush hours (6–8 PM) recorded the highest trip volume at over 3M trips per hour; early morning hours (3–5 AM) commanded the highest average fares (~$15–16) despite the lowest volume; credit card overtook cash as the dominant payment method with 30.8M credit card trips versus 16.1M cash trips, growing significantly from 2015 to 2016; average credit card tip was $1.77 per trip while cash trips recorded $0.00 as cash gratuities are not captured in the dataset; and the highest pickup density is concentrated in the longitude band -73.98 to -73.99 based on the location_summary coordinate clustering.

Future directions include adding Apache Kafka for real-time trip streaming, expanding the dataset to cover more years, incorporating weather data correlation analysis, and deploying the dashboard to Power BI Service for online access.

References

[1] NYC Taxi and Limousine Commission — Trip Record Data. https://www.nyc.gov/site/tlc/about/tlc-trip-record-data.page

[2] https://github.com/aditto-ashir-40049/nyc-taxi-azure-pipeline

[3] Kaggle NYC Yellow Taxi Trip Data Dataset. https://www.kaggle.com/datasets/elemento/nyc-yellow-taxi-trip-data

[4] Azure Cloud Architecture Reference. https://learn.microsoft.com/en-us/azure/architecture/browse/